

**Early Participation in the Trusted Exchange Framework and Common Agreement Was
Not Associated With Lower Hospital Mortality Scores: A U.S. Hospital-Level Analysis**

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Note on Data Sources: This study used two publicly available datasets: (a) the Office of the National Coordinator for Health Information Technology's *US Hospital Participation in Health Information Networks* dataset from the Health IT Research and Analysis Program and (b) the Centers for Medicare & Medicaid Services' *Complications and Deaths - Hospital* dataset from the Provider Data Catalog. Full source citations are provided in the References.

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Abstract

Given limited evidence linking health information exchange (HIE) to mortality, this cross-sectional secondary analysis examined whether early participation in the Trusted Exchange Framework and Common Agreement (TEFCA), a national HIE initiative launched in late 2023, was associated with lower hospital mortality scores. Hospital-level data from the Office of the National Coordinator for Health Information Technology (ONC) and the Centers for Medicare & Medicaid Services (CMS) were linked for 691 hospitals with usable Hybrid Hospital-Wide Mortality (Hybrid HWM) scores, a risk-standardized all-cause mortality measure for which higher scores indicate worse performance. Hospital TEFCA participation status was classified as current, planned, or neither current nor planned. Adjusted linear regression estimated associations between TEFCA status and Hybrid HWM score, using planned participation as the reference group and adjusting for national, electronic health record (EHR) vendor, and state/regional/local network participation, CMS denominator size, state, and ONC survey year. Current participation was associated with a modestly higher Hybrid HWM score than planned participation ($\beta = 0.108$, 95% CI [0.001, 0.216], $p = .047$), but the association was not significant using robust standard errors ($p = .072$). Secondary condition-specific and sensitivity analyses showed no consistent mortality advantage for current participation. However, EHR vendor network participation was consistently associated with lower mortality scores. Overall, early TEFCA participation was not associated with measurably lower mortality, and the associations with mortality across HIE arrangements remained uncertain. Findings are preliminary, given the brief cross-sectional observation period and the use of participation as a proxy for actual HIE activity.

Hospitals increasingly depend on the timely, secure exchange of electronic protected health information (ePHI) generated outside their own facilities or electronic health record systems to support emergency care, transfers, readmissions, and post-discharge transitions. In the United States, the HITECH Act of 2009 accelerated electronic health record (EHR) adoption, but health information networks (HINs) that supported EHR-mediated ePHI exchange remained fragmented by incompatible data standards and localized legal agreements that limited the flow of information across systems (Office of the National Coordinator for Health Information Technology [ONC], 2015). The 21st Century Cures Act of 2016 sought to address these limitations by advancing health information interoperability standards and directing ONC to develop or support the nationwide exchange framework that became the Trusted Exchange Framework and Common Agreement (TEFCA) (ONC, 2026b; 21st Century Cures Act, 2016, § 4003(b)).

TEFCA was developed as a public-private “network of networks” to standardize exchange across Qualified Health Information Networks (QHINs) and bridge existing health information exchange (HIE) infrastructure nationwide, thereby creating “a universal floor for interoperability” and reducing the need for healthcare providers to join multiple networks (ONC, 2026b). Following the publication of Version 1 in January 2022, TEFCA went live with initial QHIN designations in December 2023 (ONC, 2026a). Despite this policy progress, access to exchanged information does not guarantee its use in clinical practice. In 2023, 71% of hospitals reported routine electronic access to necessary external clinical information, but only 42% reported that clinicians often used it in patient care (Gabriel et al., 2024). This gap suggests that HIE’s clinical value may depend not only on access but also on whether clinicians find it useful enough to incorporate into patient care.

At the network level, the value of HINs and broader exchange frameworks such as TEFCA may likewise depend on the breadth of adoption, because broader participation can increase access to ePHI across diverse care settings for exchange purposes such as care coordination, population health monitoring, and patient access to records (Everson et al., 2024; ONC, 2026a, 2026b). Yet TEFCA adoption remained uncertain in the months leading up to rollout. In a survey of 76 health information organizations (HIOs), which operate local, regional, or state networks for electronic HIE, 63% planned to participate in TEFCA, 6% did not, and 32% were unsure (Everson et al., 2024), as concerns about TEFCA's value, security, technical barriers, and costs may have contributed to participation uncertainty (Everson et al., 2024; Griffin, 2026; Posnack, 2025; Zen Healthcare IT, n.d.). Therefore, examining early mortality associations across TEFCA and other HIE arrangements could help stakeholders assess whether different forms of exchange infrastructure are associated with positive clinical outcomes.

Despite these adoption concerns, prior research has associated HIE with reduced duplicate testing and emergency department costs, improved medication reconciliation, and lower hospital readmissions (Dupont et al., 2022; Frisse et al., 2012; Hersh et al., 2015; Hincapie & Warholak, 2011; Menachemi et al., 2018; Sadoughi et al., 2018). However, evidence for primary clinical outcomes, such as mortality, remains limited and largely observational, with weak associations between HIE and reductions in unplanned readmissions or mortality (Dupont et al., 2022; Hersh et al., 2015). Some studies suggest that HIE participation may be associated with mortality or care efficiency: Turbow et al. (2023) found lower adjusted odds of in-hospital death among older adults with Alzheimer's disease readmitted to a different hospital sharing an HIE network with the first hospital, while Kim and Choi (2025) and Janakiraman et al. (2022) found associations between HIE and shorter emergency department stays. But such research has

examined diverse HIE arrangements and mostly identified benefits for healthcare utilization, costs, and quality of care, with few studies firmly establishing mortality outcomes or participation in a nationwide framework such as TEFCA.

The present study addressed this uncertainty gap by estimating associations between TEFCA participation and hospital mortality outcomes early in its deployment while adjusting for participation in other, potentially overlapping HIE arrangements, including national HINs, EHR vendor networks, and state, regional, or local HIOs. The public datasets used for this analysis lack direct measures of HIE volume, clinical use, and workflow integration, so network participation served as a proxy for HIE. TEFCA-participating hospitals were primarily compared with hospitals planning to participate in TEFCA during the study window, because the latter may more closely resemble participants in terms of digital maturity and interoperability orientation than hospitals without a TEFCA commitment. However, this comparison, together with adjustment for other HIN participation, yielded adjusted associations rather than estimates of the clinical effect of TEFCA-enabled exchange relative to no HIN participation or of HIE versus no HIE.

The study linked 2023–2024 ONC network participation data with approximately aligned CMS mortality outcomes, using Hybrid Hospital-Wide Mortality (Hybrid HWM) score, a 30-day risk-standardized mortality measure, as the primary hospital-level dependent clinical variable. (CMS, 2026; ONC, 2026c). Adjusted linear regression tested the hypothesis that current TEFCA participation would be associated with lower mortality scores than those of hospitals with planned participation or neither current nor planned participation, while accounting for other network participation and measured covariates. Current participation was instead associated with a slightly higher adjusted Hybrid HWM score than planned participation, and secondary analyses

showed no consistent TEFCA mortality advantage. EHR vendor network participation was associated with lower mortality scores in the primary model.

Materials and Methods

Study Design, Data Sources, and Sample Selection

This study used a cross-sectional, hospital-level observational design to estimate the association between early TEFCA participation and CMS hospital mortality outcomes. Each outcome model included one observation per hospital. ONC network participation data from 2023 and 2024 were drawn from the *US Hospital Participation in Health Information Networks* dataset, which contains annual observations from 2022–2025 based on the AHA Information Technology Supplement Survey of U.S. non-federal acute care hospitals (ONC, 2026c). CMS mortality data were obtained from the *Complications and Deaths - Hospital* dataset in the CMS Provider Data Catalog, which contains hospital-level measures of complications, patient safety, and 30-day mortality (CMS, 2026). Measure-specific measurement periods are described below.

Hybrid Hospital-Wide Mortality (Hybrid HWM) was selected as the primary outcome because it provided the broadest hospital-level mortality measure and included less pre-TEFCA time than the other available outcomes, reducing the extent to which earlier observations could dilute associations corresponding to the framework’s initial rollout. The 2023–2024 ONC observations were selected as the survey years most closely aligned with the Hybrid HWM measurement period. Within the selected 2023–2024 ONC data, each hospital appeared in only one survey year, so no cross-year deduplication or survey-year precedence rule was necessary. Because CMS provided a single value per hospital for each outcome period, rather than repeated annual observations, the study estimated differences across hospitals within the approximately

one-year analytic window rather than changes within the same hospitals over time. Figure A1 in the Appendix shows TEFCA rollout and data-source alignment. Hospitals were eligible for the primary sample if they had the required ONC observations, a valid numeric CMS certification number (CCN) that linked to a CMS Facility ID, and a non-missing Hybrid HWM score. Heart failure and pneumonia samples were constructed separately because outcome availability differed by measure.

Measures

TEFCA participation status was classified as currently participating (current), planning but not currently participating (planned), or neither currently participating nor planning to participate (neither). Planned hospitals served as the primary comparison group because they may have been more similar in HIE maturity to hospitals participating in TEFCA during the survey year than to hospitals with no TEFCA commitment. The primary outcome, a Hybrid HWM score, measures all-cause mortality within 30 days of admission among eligible Medicare Fee-for-Service inpatients aged 65–94 (Yale New Haven Health Services Corporation, Center for Outcomes Research & Evaluation [CORE], 2023; Electronic Clinical Quality Improvement Resource Center, 2024). Its measurement period in the analyzed dataset was July 1, 2023, through June 30, 2024 (CMS, 2026). Secondary outcomes, when available, were the CMS 30-day all-cause risk-standardized mortality measures for heart failure (“MORT_30_HF”) and pneumonia (“MORT_30_PN”). Both had measurement periods from July 1, 2021, through June 30, 2024 (CMS, n.d.). For all three outcomes, higher values indicated higher mortality. Hospital-level HIE indicator measures included participation in TEFCA, national HINs, EHR vendor networks, and state, regional, or local HIOs. These indicators were not mutually exclusive with one another or with TEFCA participation. Other model covariates included CMS

denominator size—the number of eligible cases or admissions in the relevant measure (CMS, 2025)—state, and ONC survey year.

Data Linkage and Processing

The ONC and CMS CSV files were linked by standardizing hospital identifiers as character strings and matching the ONC CCN field “mcnum” to the CMS “Facility ID” field. TEFCA status was derived from ONC indicators of current and planned participation using the three categories described above. Consistent with ONC coding rules, missing network participation values were treated as nonparticipation. Other network indicators were coded independently (1 = reported participation; 0 = nonparticipation). CMS outcome rows were filtered to retain Hybrid HWM for the primary analysis and heart failure and pneumonia mortality for secondary analyses, which were exploratory because their three-year CMS measurement periods included substantial pre-TEFCA outcome time. Appendix Figure A2 shows the processing pipeline.

Statistical Analysis

The primary analysis used multivariable ordinary least squares (OLS) linear regression. The model was specified as:

$$\text{Hybrid HWM}_i = \beta_0 + \beta_1(\text{Current TEFCA}_i) + \beta_2(\text{Neither current nor planned TEFCA}_i) + \beta_3(\text{National network}_i) + \beta_4(\text{EHR vendor network}_i) + \beta_5(\text{HIO}_i) + \beta_6(\log \text{Denominator}_i) + \gamma(\text{State}_i) + \delta(\text{Survey year}_i) + \varepsilon_i.$$

With planned participation as the reference category, β_1 compared hospitals with current participation to those with planned participation, whereas β_2 compared hospitals with neither planned nor current participation to those with planned participation. All coefficients represented

between-hospital differences rather than longitudinal within-hospital changes. Secondary models repeated this approach with the CMS 30-day heart failure and pneumonia mortality outcome measures. Sensitivity analyses used a 2024-only ONC subset, denominator-weighted least squares, and a sample restricted to current and planned participants. Regression diagnostics examined residual behavior, unequal variance, multicollinearity, and influential observations. Because some state fixed-effect categories contained only one hospital and thus had leverage values of 1, HC3 heteroskedasticity-robust standard errors were estimated after excluding singleton-state observations. Reproducible scripts for data cleaning, linkage, variable construction, and modeling were written in R version 4.5.1, with statistical significance assessed at $\alpha = .05$.

Results

The primary analytic sample included 691 hospitals: 16.2% were current TEFCA participants, 41.8% planned participants, and 42.0% neither current nor planned participants. Hybrid HWM scores ranged from 2 to 6, with higher scores indicating higher risk-standardized mortality. Unadjusted mean Hybrid HWM scores were 4.19 among current participants, 4.06 among planned participants, and 4.40 among hospitals with neither status. Appendix Table A1 presents full descriptive characteristics. In the adjusted primary OLS model, current TEFCA participation was associated with a 0.108-point higher Hybrid HWM score (SE = 0.055, 95% CI [0.001, 0.216], $p = .047$), but this result lost statistical significance in the HC3 sensitivity analysis excluding singleton-state observations ($\beta = 0.108$, 95% CI [-0.010, 0.227], $p = .072$). Hospitals with neither current nor planned participation also had higher scores than planned hospitals, but the difference was not statistically significant ($\beta = 0.080$, SE = 0.050, 95% CI [-0.018, 0.179], $p = .110$). Among the reported covariates, EHR vendor network participation

and log CMS denominator were associated with lower Hybrid HWM scores (both $p < .001$). Appendix Table A2 and Figure A3 present the full primary model results.

In the secondary models, current versus planned TEFCA participation was associated with slightly lower heart failure mortality scores and higher pneumonia mortality scores, but neither association was statistically significant. Hospitals with neither current nor planned participation had significantly higher pneumonia mortality scores than those with planned participation. EHR vendor network participation was again associated with lower mortality scores in both secondary models. Appendix Table A3 presents the full secondary model results.

The current-versus-planned TEFCA coefficient remained positive across all three sensitivity models. Only the denominator-weighted least squares model was statistically significant ($p < .001$), in the same score-increasing direction as the primary estimate. Diagnostics indicated no substantial multicollinearity, while evidence of unequal variance and leverage from singleton-state observations supported the robust-standard-error sensitivity analyses. In the HC3 sensitivity analysis that excluded singleton-state observations, the corresponding heart failure and pneumonia comparisons remained nonsignificant. Appendix Table A4 presents the full sensitivity results, and Figure A4 shows the alternative model variants for the primary outcome.

Discussion

This study examined whether early TEFCA participation was associated with hospital mortality after adjustment for other HIE network participation. The hypothesis was not supported: current participation was not associated with lower Hybrid HWM or condition-specific mortality scores during the available observation period. The modest positive coefficient for current TEFCA participation, however, should not be interpreted as TEFCA worsening mortality performance, because unmeasured differences between early and planned

participants may have influenced the estimate. The analysis used a single, hospital-level observation period, leaving undetermined whether participants already had higher mortality scores before TEFCA participation. Additionally, sensitivity analyses preserved the positive direction of the current-versus-planned estimate, but its statistical significance was not consistent, as the association was nonsignificant in the HC3 analysis and in two of the three alternative tested models (Appendix Table A4). The inverse EHR vendor network and CMS denominator associations were secondary adjustment findings that may reflect digital maturity, hospital volume, size, or case composition rather than causal effects.

The findings should also be interpreted in the context of TEFCA's early rollout. TEFCA went live near the end of 2023, and the CMS outcome measurement periods may not have allowed sufficient exposure to TEFCA-related exchange to influence mortality outcomes. Recent federal actions and reported TEFCA milestones further suggest that the framework continued to mature after the study period (U.S. Department of Health and Human Services, 2026). Network participation was also a problematic proxy for meaningful HIE use because it reflected facility-level participation rather than actual HIE activity or clinical integration. Because hospitals could participate in multiple HINs, the model did not isolate HIE participation versus no HIE, but instead estimated adjusted associations for specific network indicators. Finally, the CMS mortality measures apply to specific eligible populations and may not capture potential benefits in other patient groups or clinical settings. Residual confounding may remain because the available data lacked teaching status, ownership, rurality, bed size, referral role, and organizational resources, while risk standardization may not have fully captured case mix or baseline performance.

Prior studies have associated HIE with operational and utilization benefits, but evidence for primary clinical outcomes such as mortality remains less conclusive (Dupont et al., 2022; Hersh et al., 2015; Menachemi et al., 2018; Sadoughi et al., 2018). The present findings did not resolve this uncertainty, which remains for HIE generally and extends to TEFCA, a framework designed to enable broad improvements in care delivery and health outcomes by connecting disparate HIE networks nationally. EHR vendor network participation was consistently associated with lower mortality scores after adjustment for other network indicators and covariates, an association that may reflect interoperability infrastructure more closely integrated with hospitals' clinical systems, but also unmeasured characteristics such as digital maturity or organizational resources.

Future longitudinal studies should examine repeated hospital observations after TEFCA has matured, incorporate hospital structural characteristics, and measure HIE volume rather than participation alone. These steps would help distinguish changes following participation from preexisting between-hospital differences and reduce residual confounding. Because this cross-sectional analysis examined TEFCA during its early rollout, the findings should be viewed as preliminary, and the secondary associations involving EHR vendor networks and CMS denominator size are hypothesis-generating. These findings do not rule out clinical benefits as HIE implementations mature and exchanged information is increasingly incorporated into clinical care.

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Appendix

Tables and Figures

Table A1

Descriptive Characteristics of the Primary Analytic Sample by TEFCA Participation Status

Characteristic	Planned	Current	Neither
Hospitals, n	289	112	290
ONC year 2023, n (%)	157 (54.3%)	46 (41.1%)	157 (54.1%)
ONC year 2024, n (%)	132 (45.7%)	66 (58.9%)	133 (45.9%)
States represented, n	47	32	44
National network participation, n (%)	223 (77.2%)	85 (75.9%)	118 (40.7%)
EHR vendor network participation, n (%)	199 (68.9%)	71 (63.4%)	67 (23.1%)
State/regional/local HIO participation, n (%)	262 (90.7%)	104 (92.9%)	195 (67.2%)
CMS denominator, median [IQR]	637 [171, 1392]	608 [110, 1591]	142 [65, 494]
Hybrid HWM score, mean (SD)	4.06 (0.51)	4.19 (0.48)	4.40 (0.46)

Note. Planned = planning to participate in TEFCA but not currently participating; Current = currently participating in TEFCA; Neither = neither currently participating nor planning to participate. Percentages are column percentages. TEFCA = Trusted Exchange Framework and Common Agreement; ONC = Office of the National Coordinator for Health Information Technology; EHR = electronic health record; HIO = health information organization; CMS = Centers for Medicare & Medicaid Services; HWM = Hybrid Hospital-Wide Mortality; IQR = interquartile range; SD = standard deviation. CMS denominator refers to the number of eligible

hospital cases/admissions included in the CMS Hybrid HWM measure. Higher Hybrid HWM scores indicate higher risk-standardized mortality.

Table A2

Primary Linear Regression Model Predicting Hybrid HWM Score

Predictor	Estimate	SE	95% CI	p-value
Current vs planned	0.108	0.055	[0.001, 0.216]	.047*
Neither vs planned	0.080	0.050	[-0.018, 0.179]	.110
National network	-0.026	0.042	[-0.108, 0.057]	.544
EHR vendor network	-0.186	0.045	[-0.274, -0.099]	< .001**
State/regional/local HIO	-0.026	0.051	[-0.126, 0.073]	.603
Log CMS denominator	-0.075	0.015	[-0.103, -0.046]	< .001**

Note. Planned TEFCA participation was the reference category. The model adjusted for national network participation, EHR vendor network participation, HIO participation, log CMS denominator size, state fixed effects, and ONC survey-year fixed effects. State and year fixed effects are not shown. Log CMS denominator is the natural logarithm of the number of eligible hospital cases/admissions included in the CMS Hybrid HWM measure. Higher Hybrid HWM scores indicate higher risk-standardized mortality. N = 691. * $p < .05$. ** $p < .001$.

Table A3*Secondary Linear Regression Models Predicting Heart Failure and Pneumonia Mortality Scores*

Predictor	Estimate	SE	95% CI	p-value
Heart failure mortality				
Current vs planned	-0.024	0.265	[-0.545, 0.497]	.927
Neither vs planned	0.439	0.261	[-0.073, 0.951]	.093
National network	0.023	0.213	[-0.395, 0.441]	.914
EHR vendor network	-0.462	0.226	[-0.906, -0.018]	.041*
State/regional/local HIO	0.374	0.268	[-0.154, 0.901]	.164
Log CMS denominator	-0.386	0.094	[-0.571, -0.202]	< .001**
Pneumonia mortality				
Current vs planned	0.479	0.302	[-0.115, 1.073]	.114
Neither vs planned	0.588	0.281	[0.037, 1.140]	.037*
National network	-0.125	0.239	[-0.595, 0.345]	.600
EHR vendor network	-0.744	0.255	[-1.245, -0.243]	.004*
State/regional/local HIO	0.172	0.287	[-0.392, 0.735]	.550
Log CMS denominator	-0.224	0.116	[-0.452, 0.004]	.054

Note. Planned TEFCA participation was the reference category. Each model adjusted for national network participation, EHR vendor network participation, HIO participation, log CMS denominator size (the natural logarithm of the number of eligible hospital cases/admissions included in each CMS condition-specific mortality measure), state fixed effects, and ONC survey-year fixed effects. State and year fixed effects are not shown. Heart failure model: N =

524, $R^2 = .227$, adjusted $R^2 = .136$. Pneumonia model: $N = 630$, $R^2 = .202$, adjusted $R^2 = .126$.

Higher scores indicate higher CMS condition-specific 30-day mortality. $*p < .05$. $**p < .001$.

Table A4

Sensitivity Analyses for the Current TEFCA Versus Planned TEFCA Comparison

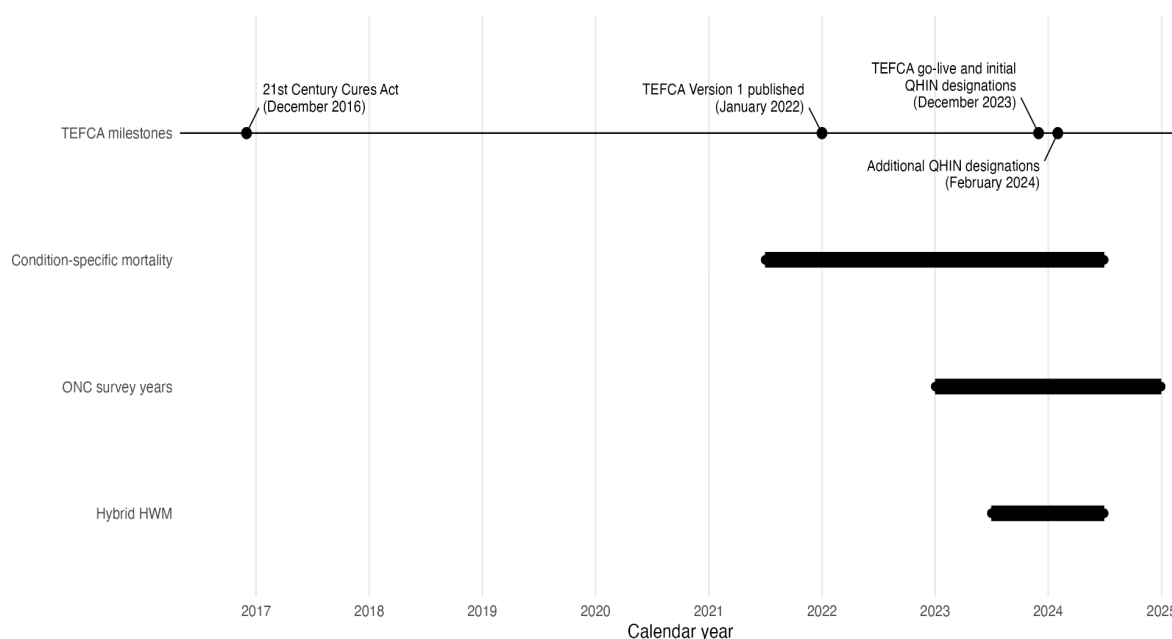
Model or outcome	N	Estimate	SE	95% CI	p-value	R ²	Adjusted R ²
Panel A. Primary-Outcome Sensitivity Models							
Main OLS model	691	0.108	0.055	[0.001, 0.216]	.047*	0.321	0.262
2024-only model	331	0.128	0.078	[-0.027, 0.282]	.105	0.365	0.247
Weighted least squares model	691	0.179	0.053	[0.074, 0.283]	< .001**	0.463	0.417
Current-and-planned-only model	401	0.113	0.058	[-0.002, 0.227]	.054	0.358	0.260
Panel B. HC3 Robust-Standard-Error Analyses Excluding Singleton-State Observations							
Hybrid HWM	690	0.108	0.060	[-0.010, 0.227]	.072		
Heart failure mortality	521	-0.024	0.282	[-0.578, 0.530]	.932		
Pneumonia mortality	629	0.479	0.318	[-0.146, 1.104]	.133		

Note. Estimates are adjusted coefficients for current TEFCA participation relative to planned TEFCA participation. Positive estimates indicate higher mortality scores among hospitals currently participating. All models adjusted for participation in national networks, EHR vendor networks, and state/regional/local HIOs, log CMS denominator, state, and ONC survey year, except that the 2024-only model omitted survey-year fixed effects. CMS denominator refers to

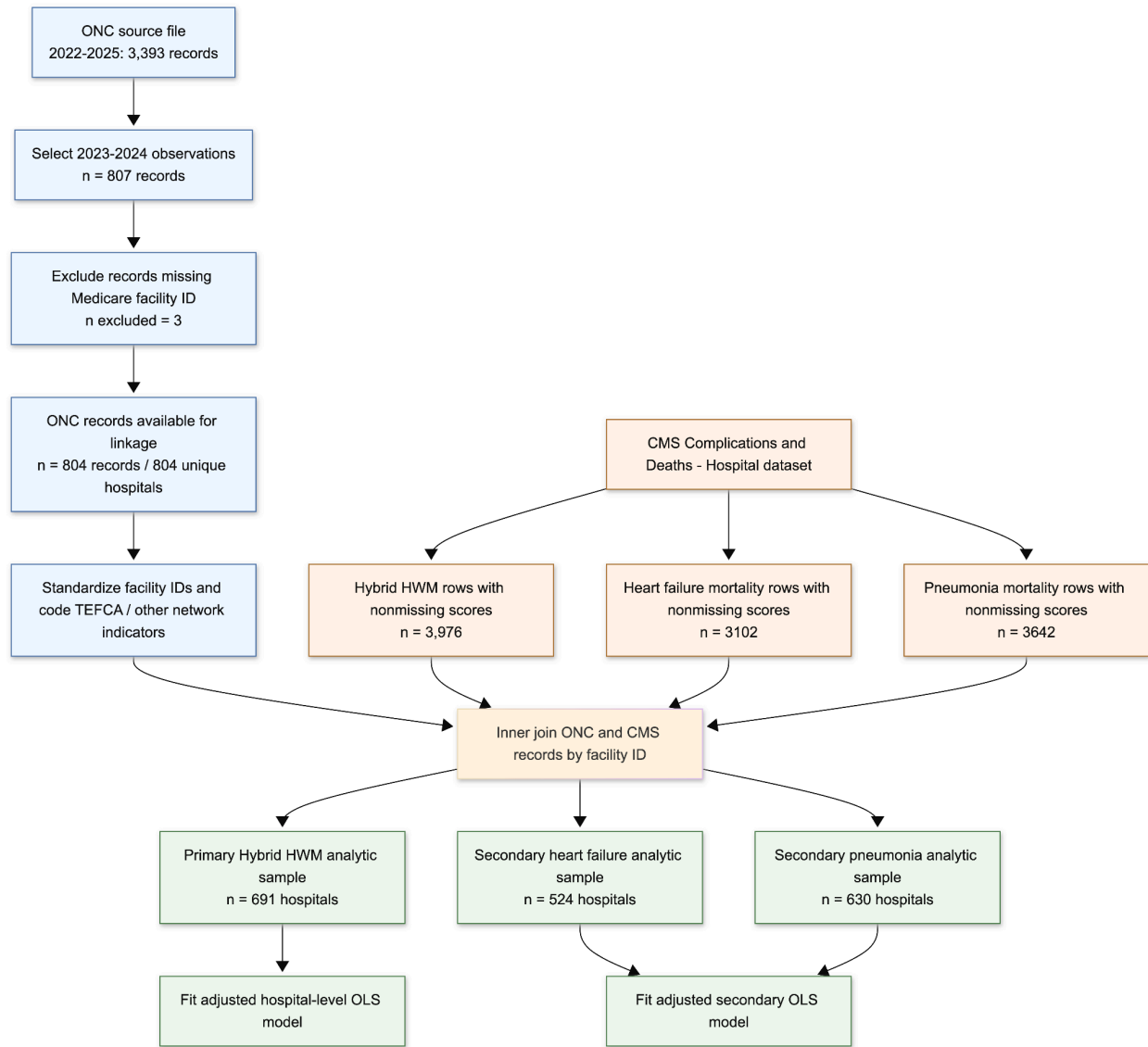
the number of eligible hospital cases or admissions included in the relevant measure. The weighted least squares model used the untransformed CMS denominator as the analytic weight, and the current-and-planned-only model excluded hospitals with neither current nor planned TEFCA participation. Panel B reports HC3 heteroskedasticity-robust standard errors after excluding states represented by one hospital in the relevant analytic sample. The excluded states were DC for Hybrid HWM and pneumonia mortality, and DC, ND, and NM for heart failure mortality. $*p < .05$. $**p < .001$.

Figure A1

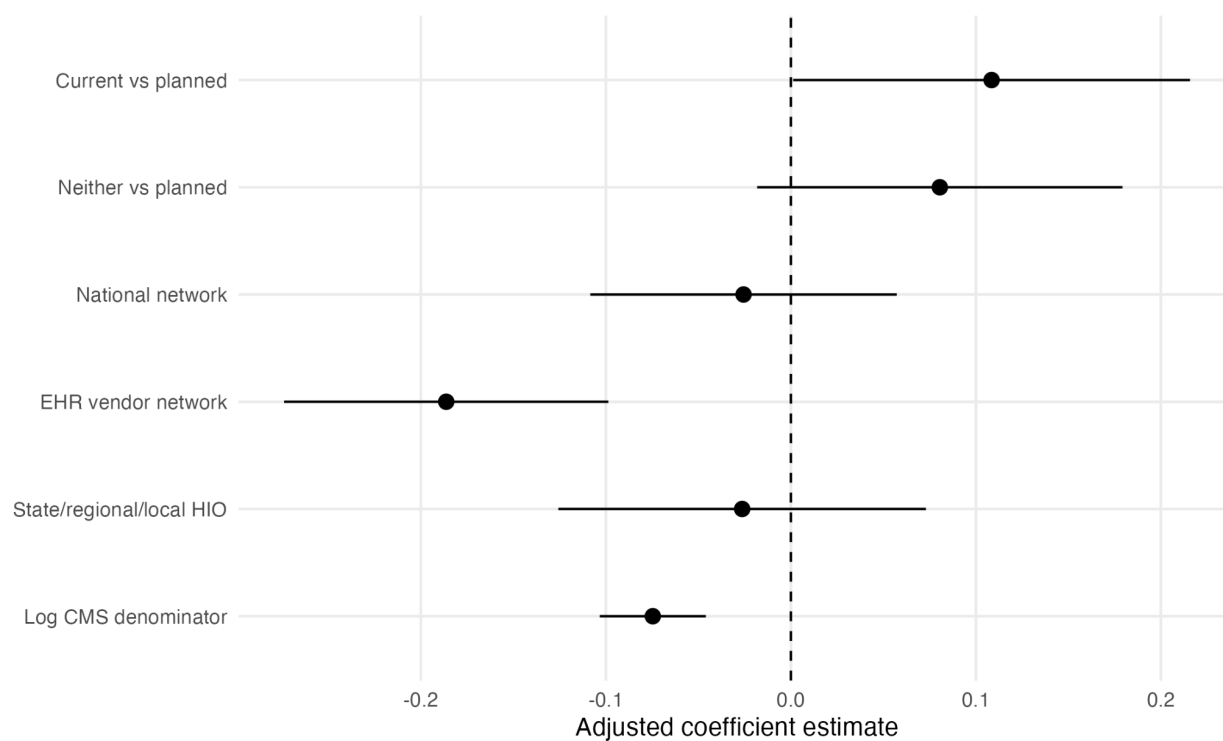
TEFCA and Data Alignment Timeline



Note. The timeline shows the relationship between TEFCA rollout milestones and CMS outcome measurement periods, illustrating why the study used a cross-sectional hospital-level design centered on the approximately one-year Hybrid HWM measurement window and why findings are interpreted as temporally informed associations rather than definitive causal effects.

Figure A2*Data Processing Pipeline*

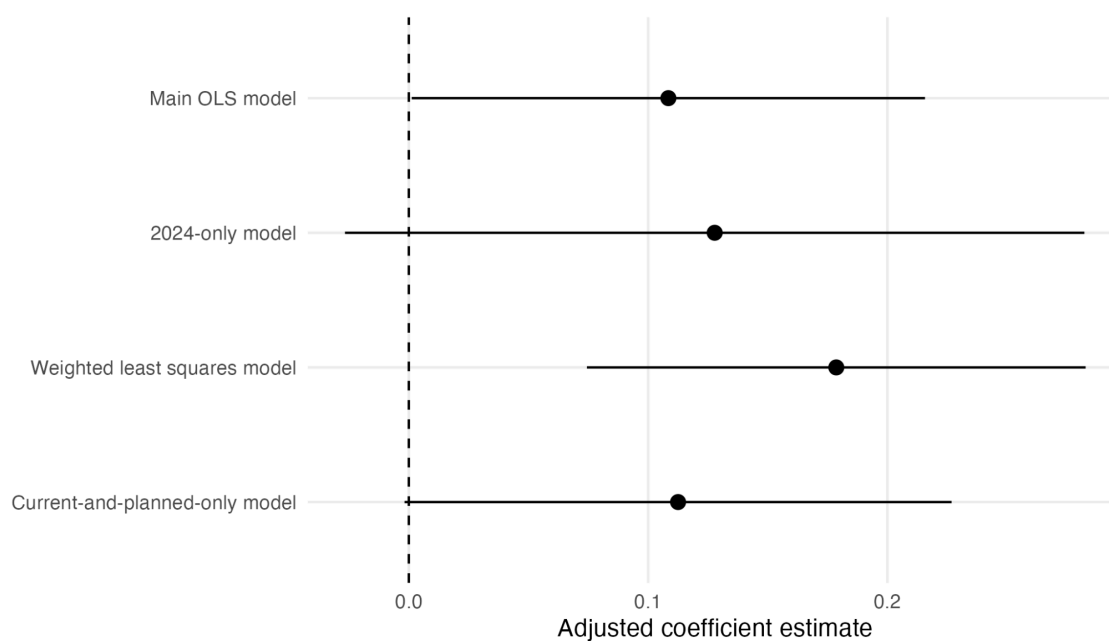
Note. Data processing workflow for linking ONC hospital network participation data to CMS hospital mortality outcomes and for cleaning the merged dataset.

Figure A3*Primary Model Coefficient Estimates for Hybrid HWM Score*

Note. Points represent adjusted OLS coefficient estimates; horizontal bars represent 95% confidence intervals. Planned TEFCA participation was the reference category. State and ONC survey-year fixed effects were included but are not shown. Log CMS denominator is the natural logarithm of the number of eligible hospital cases/admissions included in the CMS Hybrid HWM measure.

Figure A4

Sensitivity Analyses for the Current TEFCA Versus Planned TEFCA Estimate



Note. Points represent adjusted coefficients for current TEFCA participation relative to planned participation; horizontal bars represent 95% confidence intervals. The weighted least squares model used the CMS denominator as the analytic weight. The 2024-only model omitted year fixed effects because only one ONC survey year was included.

Generative AI Acknowledgment

- **Name of the AI tools used**

- ChatGPT

- **Version and extension numbers:** GPT-5.5 Thinking and GPT-5.6 Sol
- **Manufacturer:** OpenAI
- **Date of use:** June-July, 2026
- **Description of use:** ChatGPT was used to support the development and revision of this research paper on TEFCA participation. It was used to help plan and implement a reasonably scoped statistical analysis of the datasets. I also solicited feedback on drafts of each major section and the Materials and Methods subheadings, and used this tool to explore possible wording and organization options for the Materials and Methods, Results, and Discussion sections, as well as the title and abstract. Used alongside Grammarly, I often chose to implement suggestions for clearer word choice and to correct ideas I drafted that did not align with the scope or strength of the analysis. I reviewed, selected, revised, and made final decisions on all suggested claims, data analysis decisions, and final wording. I also used this tool to ask for clarification on the technical underpinnings of the chosen statistical model, for example, the limitations on causal inference from the model's adjusted estimates.

- Gemini

- **Version and extension numbers:** 1.5 Pro
- **Manufacturer:** Google
- **Date of use:** June 18, 2026
- **Description of use:** Gemini was used to refine sentence structure, specifically in the first paragraph, where the HITECH Act's limitations are mentioned. It was also used to retrieve and format an appropriate source for that claim in APA 7th edition format. It was also used to understand whether TEFCA participation is mandatory for U.S. HIOs.

- Grammarly

- **Version and extension numbers:** Grammarly 1.173.1.0; WebUI 2.16.13
- **Manufacturer:** Grammarly, Inc.
- **Date of use:** June–July, 2026
- **Description of use:** Grammarly was used as an integrated desktop writing assistant. It provided suggestions on syntax, grammar, punctuation, and word choice during drafting. I reviewed each suggestion before accepting or rejecting it, and I made final wording decisions.